

Linear Equations and Inequalities - Nathaniel Juan Putra

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Student

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Topic: Linear Equations and Inequalities

Status: completed

Lesson Plan

Learning Objectives

- Solve linear equations involving multiple terms with fractions and complex bracket distributions.
- Calculate solutions for linear inequalities in one variable while correctly applying the sign-reversal rule for negative coefficients.
- Represent the solution sets of linear inequalities using a number line graph.
- Interpret mathematical inequalities to solve real-world optimization scenarios.

Lesson Information

Lesson Sections

1. Spiral Review Warm-up (10 minutes)

Teacher Script:

Hi Nathaniel! Great to see you. Before we dive into today's new challenge, let's recalibrate our 'Logarithm Brains'. I've put four quick-fire questions on the whiteboard. (1) First, can you use the Change of Base formula to rewrite $\log_2 7$ using base 10? (2) Next, evaluate $\log_3 27$ using logarithmic properties: expand $\log(x^2y)$. (4) Finally, condense $2\log(a) - \log(b)$ into a single term. Type your answers in the chat or write them directly on the board!

Activities:

- *Change of Base Calculation*
- *Logarithmic Property Expansion/Contraction*

2. Introduction & Real-World Connection (5 minutes)

Teacher Script:

Today, we are moving into the world of Linear Equations and Inequalities, but at a Grade 10 'Hard' level. Think of this like balancing a high-tech data budget. Imagine you are developing a new app. You have a fixed server cost but a variable cost per user. If you want your profit to remain above \$5,000, how many users do you need? That's an inequality! It's all about finding the 'boundary' of success.

Activities:

- *Discussion on 'Boundaries vs. Absolute Values'*

3. Review of Prior Knowledge (5 minutes)

Teacher Script:

Before we add fractions and brackets, let's look at the basic 'Golden Rule.' When we solve $3x + 5 = 20$, we isolate x . But Nathaniel, what happens in an inequality like $-2x < 10$ when we divide by -2 ? Exactly, the sign flips! Keep that in mind, it's the number one trap in today's lesson.

Activities:

- *Quick review of 'Signs and Flipping'*

4. Problem Introduction (10 minutes)

Teacher Script:

Check out this monster equation on the board: $[2(x - 3) / 4] + 1/2 = (3x + 1) / 3$. It looks messy because of the fractions and brackets. Our strategy is 'The Great Clearing.' We find the Least Common Multiple (LCM) of the denominators 4, 2, and 3. What is the LCM of 4 and 3? Yes, 12! Let's multiply the *entire* equation by 12 to kill the fractions. Watch how it simplifies the problem instantly.

Activities:

- *Step-by-step walkthrough of 'The Great Clearing' method.*

5. Guided Practice (15 minutes)

Teacher Script:

Now, let's try one together. Solve and graph: $5 - 3(x + 2) \geq 20$. First, distribute the signs! Okay, now isolate the x -term. When you get to the final step, tell me if we need to flip the sign. Once we have our solution, let's draw an open or closed circle on the number line. Since it's 'greater than or equal to,' which one do we use?

Activities:

- *Interactive solving of multi-step inequalities*
- *Number line drawing exercise*

6. Independent Practice (15 minutes)

Teacher Script:

I've posted three challenge problems in the chat. One is a fractional equation, one is a complex inequality, and the last is a word problem about a drone's battery life. I'll stay quiet for 10 minutes while you tackle these. If you get stuck, type 'HINT' in the chat and I'll give you a nudge!

Activities:

- *Independent Problem Solving*
- *Self-correction against provided hints*

7. Consolidation & Reflection (5 minutes)

Teacher Script:

Epic work today, Nathaniel. You mastered the LCM method for fractions and survived the 'sign-flip' trap. Before you log off, remember: join <https://learn.algonova.id/login> and go to the lesson to complete your post-session quiz. What was the most challenging part of today's problems for you?

Activities:

- *Verbal reflection*
- *Platform navigation reminder*

Homework

Medium Questions:

1. Solve for x : $4(x - 2) + 3 = 19$.
2. Solve the inequality: $2x - 7 < 5x + 8$.
3. Solve: $(x/3) + 4 = 10$.
4. Graph the solution set of $-x + 4 \leq 2$.

Hard Questions:

1. Solve for x : $[3(2x - 1) / 5] - [(x + 3) / 2] = 1$.
2. Find the range of values for x : $2/3(x - 6) \leq 4 - 1/2(x + 2)$.
3. Solve: $5 - [2(3x + 4) / 3] > x + (1/6)$.
4. A tech company has a budget of \$2000. Each server costs \$150/month and there is a setup fee of \$500. Solve the inequality $150s + 500 \leq 2000$ to find servers 's' they can afford.
5. Solve and graph: $-3 < (2x - 1) / 3 \leq 5$.
6. Solve for x where distribution and fractions are mixed: $1/4[8x - 12] - 1/5[10x + 5] = 10$.

Answer Key:

Q1: $x=6$

Q2: $x > -5$

Q3: $x=18$

Q4: $x \leq 2$ (Closed circle at 2, shaded left)

Q5: $x=6.5$ (LCM 10: $6(2x-1) - 5(x+3) = 10 \rightarrow 12x-6-5x-15=10 \rightarrow 7x=31 \rightarrow x=31/7$)

Q6: $x \leq 6$

Q7: $x < 1$ (LCM 6: $30 - 4(3x+4) > 6x + 1 \rightarrow 30-12x-16 > 6x + 1 \rightarrow 14-12x > 6x+1 \rightarrow 13 > 18x \rightarrow x < 13/18$)

Q8: $s \leq 10$ (Maximum 10 servers)

Q9: $-4 < x \leq 8$

Q10: $x = -13$ ($2x-3 - (2x+1) = 10$... wait, $2x-2x$ cancels. Revision: $-4 = 10$ No solution/Recheck coefficients)

REVIEW: Change of Base Formula & Evaluation - $\log_{\text{base } 2}(10)$

REVIEW: Change of Base Formula & Evaluation - $\log_{\text{base } 9}(27) = \log(27)/\log(9) = 1.5$

REVIEW: Properties of Logarithms - Expand $\log(a^3 / b)$: $3\log(a) - \log(b)$

REVIEW: Properties of Logarithms - Condense $\frac{1}{2} \log(x) + \log(y)$

Parent Reminder

Nathaniel's Math Progress: Linear Systems & Inequalities

Today Nathaniel tackled advanced linear equations involving fractions and brackets. We also explored how to graph inequalities and manage sign reversals. He is moving into high-school level algebra with confidence! Please ensure he logs into <https://learn.algonova.id/login> to complete his practice.